MUSCULOSKELETAL INFECTIONS

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OBJECTIVES

1. Describe the differences in pathogenesis, clinical features and treatment of osteomyelitis in the pediatric population and in adults

2. Recognize the clinical features of other bone infections seen in children: Brodie's Abscess and Pott's Puffy .

3. Discuss the pathogenesis, clinical features and treatment of septic arthritis in children and adults .

4. Describe the pathogenesis, clinical features, and treatment of Gonococcal septic arthritis and Disseminated GC

5. Discuss the pathogenesis, clinical features and treatment of pyomyositis

6. Describe the pathogenesis, clinical features and treatment of gas gangrene/nec. fasc.

7. Describe the pathogenesis, clinical features and treatment of Tetanus

8. Describe the pathogenesis, clinical features and treatment of Trichinosis

9. Describe the pathogenesis, clinical features and treatment of Cysticercosis

10. Describe the pathogenesis, clinical features and treatment of Pyomyositis

TABLET OF CONTENT

OSTEOMYELITIS

*An infection of the bone caused by pyogenic bacteria

*Two mechanism of infection

I. Hematogenous spread: Via the blood from distant sites
II. Non-hematogenous spread:

- Contiguous spread: via invasion from nearby infected skin and soft tissue .
- Direct inoculation: via penetrating traumatic or surgical wound.

Pyogenic: pus producing
Pyogenic bacteria:
Staphylococcus aureus, Streptococcus pyogenes, Escherichia coli, Klebsiella spp, Proteus spp, and Pseudomonas spp

OSTEOMYELITIS

Pus accumulates in the medullary space->

Intraosseous pressure is increased->

Impaired blood flow->

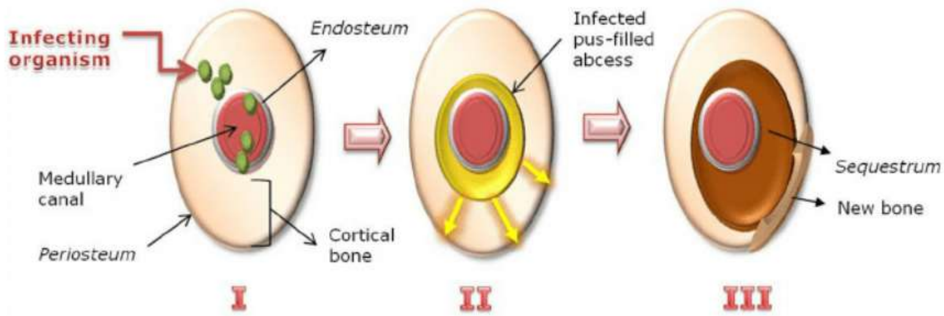
Results in ischemic necrosis and the destruction of bone

ACUTE HEMATOGENOUS OSTEOMYELITIS *2-4 WEEKS AFTER ONSET OF SX.*

	Children	Adult
Hematogenous osteomyelitis as primary route of infection	**Most common**	~20% (elderly and IV drug users).
Common Location	(**Long bone)** **Tibia, femur, and humerus**	Vertebral bodies
Incidence	25% of pts <2 years old 50% of pts <5 years old 2X more in males	
Risk Factors	Sickle cell disease Immunodeficiency disorders Orthopedic hardware Indwelling catheters	

-Variation is due to changes in the vascular anatomy.

PATHOGENESIS



Host susceptibility is increased by trauma, ischemia, and foreign bodies
Phagocytes attempting to limit infection release enzymes that lyse the bone
Vascular channels become clogged with pus, impairing blood flow
Pus eventually breaks through the cortex, forming sub-periosteal or soft tissue abscesses.
The elevated periosteum deposits new bone

In diagram:
I - A large inoculum of bacteria reaches the medullary canal
II - (Acute state) Pus resulting from inflammatory response spreads into vascular channels
III - (Chronic state) Vascular channels are compressed and obliterated by the inflammatory process, and the resulting ischemia also contributes to bone necrosis.

New bone being formed due to osteomyelitis

Moth eaten + new layer of bone.

OSTEOMYELITIS (PEDIATRICS)

ACUTE HEMATOGENOUS OSTEOMYELITIS CLINICAL PRESENTATION :

	Children
Symptoms	I. Initially nonspecific (malaise, low grade fever, etc.). *Fever of 100.4 F (40-80%) *Localized pain (55-95%) *Decreased mobility (50-84%) II. Pain -Infants and toddlers may have progressive limp -Older Children with vertebral osteomyelitis may appear toxic and very febrile with back pain.
Culture	I. If positive : A hx of antecedent trauma Erythema, swelling, or cellulitis over the infected bone A higher white blood cell count A shorter duration of symptoms II. However, up to **20-50% of cultures are negative. S. aureus is still primary suspect (50% is MRSA).**

●Constitutional symptoms (irritability, decreased appetite, aches activity), with or without fever
●Focal symptoms and signs of bone inflammation (eg, warmth, swelling, point tenderness) that typically progress over several days to a week; symptoms and signs in infants and small children may be poorly localized
●Limitation of function (limited use of an extremity; limp; refusal to walk, crawl, sit, or bear weight)

ACUTE HEMATOGENOUS OSTEOMYELITIS DIAGNOSIS:

	Children
Labs	
WBC	Elevated WBC in only 35% (65% WNL)
Erythrocyte sedimentation rate (ESR)	Elevated in 92% ESR (>20 mm/h)
Serum C-reactive protein (CRP)	Elevated in 98% CRP (>10 to 20 mg/L)
Imaging	
X-ray	Varying sensitivity with region and age. Can show evidence of periosteal new bone formation
MRI	More sensitive , recommended especially in older children if x-ray is inconclusive.

ACUTE HEMATOGENOUS OSTEOMYELITIS DIAGNOSIS: IMAGING

A. Evidence of periosteal new bone formation (periosteal reaction)

ACUTE HEMATOGENOUS OSTEOMYELITIS TREATMENT:

	Children
	Initially empiric antibiotics. Pending tissue/aspirate cultures.
Neonates	**S. aureus**, gram neg bacilli (E.coli), and group B strep Tx: Third generation cephalosporin + vancomycin
Older infants and children	**S. aureus** and group A strep Tx: Vancomycin
Sickle cell	Salmonella Tx: Third generation cephalosporin

Even if the initial x-rays and cultures are negative, suspected osteomyelitis must be treated

Duration:

-IV antibiotics for ~2 weeks followed by

-Oral antibiotics for 4-6 weeks (ESR levels monitored)

-Less than 3 weeks can lead to tx failure

-Chronic osteomyelitis is rare in children

In the pre-antibiotic era, 20% of children infected with S. aureus osteomyelitis died
Aspirates should be used for culture unless surgery is to be immediate
Consider Vancomycin in Sickle Cell as it could be polymicrobial

ACUTE HEMATOGENOUS OSTEOMYELITIS MONITORING:

	Children
Clinical Signs	Assess activity level and appetite Routine physical exam of affected region (s)
Labs	Trend ESR and CRP levels **-CRP** returns to normal faster than ESR (10 days vs. 30 days)
Imaging	X-rays not as sensitive: areas of devitalized bone (sequestra) are evident with 10-21 days after symptoms onsent.

Very UNLIKELY that any abnormal radiological findings will be seen for the first 7 days
More sensitive in younger children.
Shoulder still be treated if imaging is normal.

ACUTE HEMATOGENOUS OSTEOMYELITIS TREATMENT:

Children
<u>**Indications for Surgery:**</u>
-Subperiosteal and soft tissue abscesses must be drained.
-Sequestra must be removed.
-Contiguous infectious foci should be debrided.
-Children who are not improving on drug therapy sometimes require additional procedures to eliminate collections of pus and necrotic bone.

Limited response to therapy.

BRODIE'S ABSCESS:
RARE, SUBACUTE HEAMATOGENOUS OSTEOMYELITIS.

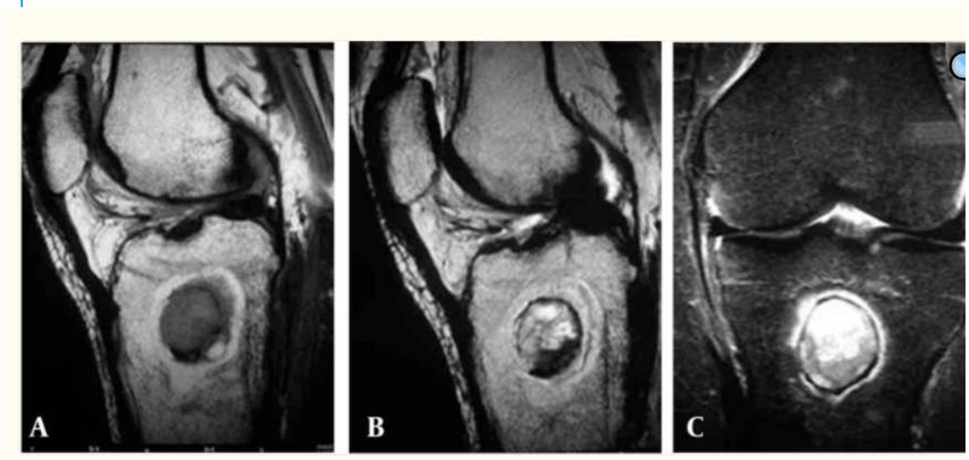
Demographic	2-15 year old most common type of subacute osteomyelitis.
Location	Distal tibia is the most common site
Symptoms	Difficult to diagnose. Onset is insidious. Mild or no fever, dull pain, and periosteal elevation. May mimic a tumor
Diagnostics	Labs are usually normal. No x-ray findings for 2 weeks. MRI sensitive (penumbra sign). Consider biopsy.
Treatment	Surgical debridement and culture-specific antibiotics are often effective
Unique feature	virulence of the organism and the resistance of the patient are more or less evenly balanced

History:

In 1832, Sir Benjamin Brodie, a London surgeon amputated the leg of a person with years of intractable pain and found a cavity in the bone

Initial diagnosis is made based on clinical suspicion and radiographic finding

BRODIE'S ABSCESS



Penumbra sig: well-defined central intramedullary cystic lesion in the proximal tibial metaphysis

A, T1-weighted TSE sequence reveals the penumbra sign; B, T2-weighted TSE sequence reveals the penumbra sign; C, MR image STIR-TSE sequence shows the penumbra sign.

The central part of the lesion was hypointense on T1-weighted images , and hyperintense on T2-weighted image and short tau inversion recovery (STIR)-TSE sequence.

POTT'S PUFFY TUMOR: RARE INFECTION OF THE FRONTAL SINUS.

Demographic	Young adolescents
Location	Frontal bone
Symptoms Diagnostics	Tender forehead swelling associated with fever, headaches, nasal discharge, or symptoms of increased intracranial pressure CT or Brain MRI with contrast
Treatment	drainage and 6 weeks of isolate-specific antibiotics
Unique feature	#1 cause is sinonasal infection, #2 is trauma.
Complications	meningitis, epidural abscesses, and thrombophlebitis

Named after Sir Percival Pott, and English surgeon who described it in 1760
Historically, a serious complication of frontal sinusitis but rarely seen in the current antibiotic era.
Consists of subperiosteal abscess and osteomyelitis if the frontal bone

OSTEOMYELITIS (PEDIATRICS) SUMMARY

-Hematogenous route most common.

-Commonly effect long bones: tibia, femur, humerus.

- Initially nonspecific (malaise, low grade fever, etc.) and then progress to pain (limping, low back pain).

- ESR and CRP help with diagnosis and treatment response to treatment.

-S. aureus is the most common pathogen. Consider Salmonella in Sickle Cell.

<u>-Even if the initial x-rays and cultures are negative, suspected osteomyelitis must be treated empirically: usually third generation cephalosporin + vancomycin.</u> <u>**Treatment can be completely curative.**</u>

- Brodie abscess: most common form of subacute hematogenous osteomyelitis usually in distal tibia. Labs and clinical symptoms does no help with diagnosis, MRI most sensitive (penumbra sign)

-Pott's Puffy: rare frontal subperiosteal abscess and osteomyelitis in the frontal bone from sinonasal infections. CT/MRI brian best way to confirm diagnosis.

OSTEOMYELITIS (ADULTS)

-Most commonly arises from open fractures, diabetic foot infections, or as a result of surgical treatment of closed injuries
-The term “cure” is not used with adults because infection is likely to re-surface years later if there is a new trauma to the area or if the patient becomes immuno-compromised.

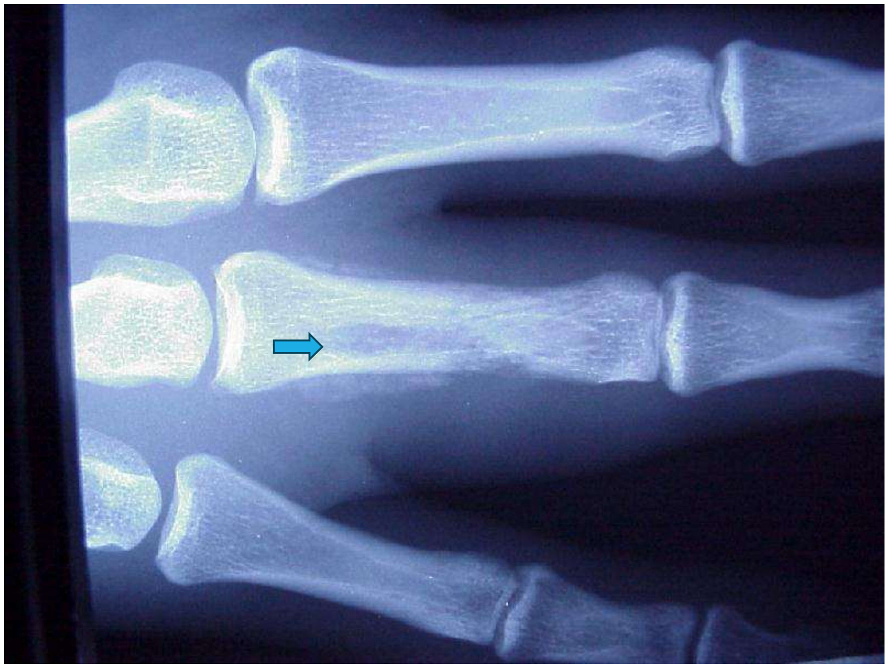
Goal of treatment in adults is to arrest the disease, rather than to cure it

OSTEOMYELITIS IN ADULTS: HEMATOGENOUS SPREAD

	Adults
Hematogenous (20% of adult osteomyelitis)	**-95% is single organism and in half of those is Staph aureus.** -Secondary hematogenous osteomyelitis more common (reactivation of an earlier infection that was only suppressed). -IV drug user: Pseudomonas and Serratia (can be in vertebrae, sacroiliac, sternoclavicular, or pubic joints) -Sickle cell: Salmonella.
Presentation	-1-3 months of dull pain (50% in lumbar region, 35% in thoracic region). Can present as atypical chest/abdominal pain due to nerve root irritations **(referred pain!)** -Minimal constitutional symptoms (DIFFERENT THAN CHILDREN).

OSTEOMYELITIS IN ADULTS: HEMATOGENOUS SPREAD

	Adults
Diagnostics:	Clinically with support from labs (ESR/CRP: sensitive) and imaging (MRI best for early, X-ray good for chronic) Bone biopsy and microbial cultures offer definitive diagnosis.
Treatment	Antibiotics and multiple surgical procedures*



Middle finger has
-interruption of the trabeculae
-erosion of bone
-periosteal new bone formation (periosteal reaction)

OSTEOMYELITIS IN ADULTS: CONTIGUOUS SPREAD.

Contiguous (80% of adult osteomyelitis)	-Direct inoculation from adjacent structures. -Example: hands after punching a tooth "fight bite". - Risk Factors: hardware, prosthetics, open fractures. - Usually polymicrobial but **methicillin-resistant *S. aureus* (MRSA), is the most common pathogen**. ☆
Presentation	Symptoms ~1 month after trauma /procedure.
Unique population	Vascular insufficiency i.e diabetic patients -Small bones of feet . -Decreased sensation. -Polymicrobial (Coagulase + and – Staph, Strep, Enterococcus, gram -. Anaerobes) - Limited diagnosis - Goal is to suppress infection - May need resection or amputation

-Contigous:
Caused by direct inoculation of bone at the time of trauma, spread from nearby soft tissue infection, or spread via nosocomial contamination during surgery
Common contributing factors include surgical hardware, prosthetic devices, and open fractures
Bone infection becomes apparent a month or more after the initial trauma or procedure and has become chronic
Early symptoms are commonly attributed to the original trauma or surgery
-Vascular:
Making a diagnosis can be challenging
Pts often present with seemingly unrelated problems: ingrown toenail, cellulitis, foot ulcer
Radiographic change is a late finding
Goal of treatment is to suppress the infection and allow the limb to survive
Resection or amputation of the area affected is almost always ultimately necessary

CHRONIC OSTEOMYELITIS IN ADULTS

Causes	Hematogenous or contiguous spread
Location	Return of symptoms at the old site of osteomyelitis
Presentation	Chronic pain and drainage, sometimes a low-grade fever
Diagnostics	-Sed rate is usually elevated -WBC is usually normal
Treatment	Limited efficacy of empiric antibiotic. Surgery: debridement, dead space management, soft tissue coverage, restoration of blood supply and stabilization
Complications	Vascular thrombosis, bone necrosis, formation of sequestra (devascularized bone).

Stopping the infection is difficult when the surrounding soft tissue is poor.

As an infection becomes chronic, the increase in intramedullary pressure causes the periosteum to become stripped from the osteum, leading to vascular thrombosis
Bone necrosis follows due to loss of blood supply.
Large fragments of devascularized bone, called sequestra, are the result
In time, pus breaks through the cortex, and forms subperiosteal abscesses and the elevated periosteum deposits new bone, called involcrum, around the sequestrum
Openings in the involcrum allow debris and exudate (pus) to pass from the sequestrum via sinus tracts to the skin

CHRONIC OSTEOMYELITIS

Bone destruction

OSTEOMYELITIS (ADULTS) SUMMARY

-Contiguous (80% of adult osteomyelitis) **(different from children).** **Usually polymicrobial, but mostly from MRSA** **(similar in children).** Can be after a surgery/procedure.

-Hematogenous (20% of adult osteomyelitis), usually caused by S. Aureus **similar in children).** IF IV drug user, think Pseudomonas and Serratia. IF sickle cell, think Salmonella (same in children).

-Clinically present as dull pain and **minimal constitutional symptoms (different from children).**

- Bone biopsy and microbial cultures offer definitive diagnosis (similar in children).

- Even with antibiotics and multiple surgical procedures, treatment is likely suppressive not 100% curative **(different from children).** especially in vascular insufficiency (i.e. diabetes). ** Also inspect patient with shoes/socks/and clothes off especially in high risk patient **.

-Chronic Osteomyelitis **more common in adults** . Can be hematogenous or contiguous spread. Can be reoccurrence of symptoms at an old site of osteomyelitis

-Most commonly arises from open fractures, diabetic foot infections, or as a result of surgical treatment of closed injuries

-The term “cure” is not used with adults because infection is likely to re-surface years later if there is a new trauma to the area or if the patient becomes immuno-compromised.

Goal of treatment in adults is to arrest the disease, rather than to cure it

SEPTIC ARTHRITIS

Acute bacterial arthritis is a medical emergency

The joint is going to be destroyed if not treated promptly

Cellulitis versus Trauma

SEPTIC ARTHRITIS

	Children	Adult
Causes: Usually Hematogenous Spread	-Staph aureus most common. ☆ -6mo-5yo: H. influenzae was the most common but that is no longer true due to the vaccine.* -Viral infections (measles, mumps) can also be causative agents	-Staph aureus, beta-hemolytic strep and gram-negative bacilli. -Pseudomonas if immunosuppressed, hx of IV drugs. -Gonococcal infection. **

* May still happen in other countries or unvaccinated.
** Have to examine other parts of the body (eyes, genitourinary region, joints).

SEPTIC ARTHRITIS

Risk factors	Patients with prosthetics Rheumatoid Arthritis Total joint replacements 1-4% will become infected around the time of surgery Diabetes mellitus Skin infection IV drug abuse, alcoholism Age >80 years

SEPTIC ARTHRITIS

	Children	Adult
Presentation:	-Infants: irritable, fever, not moving affected limb (usually **hips**). ☆ -Children: acute joint pain and stiffness (usually **knee** 40%)☆ -Constitutional symptoms - Joint warm, tender and swollen, with evidence of a generalized effusion	-Constitutional symptoms - Joint warm (usually **one joint and commonly the knee**), tender and swollen, with evidence of a generalized effusion. - Ask RF for gonorrhea.*

Knee 40% of the time.

Joint is warm, tender and swollen, with evidence of a generalized effusion

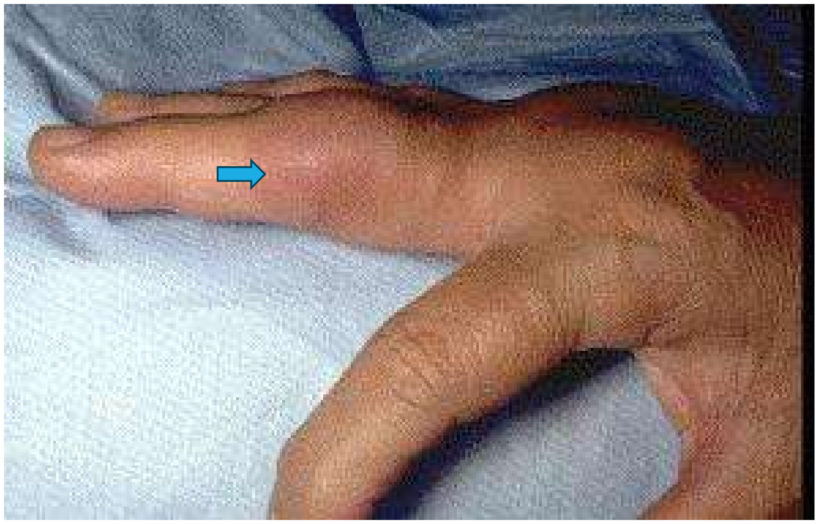
Usually there will be chills, fever, and general malaise

*Ask in children and adult (sexual history, pregnant mother did not get treated may get through birth canal).

SEPTIC ARTHRITIS, LEFT HIP JOINT

The classical position in septic hip- the affected leg is held rigidly in a position **of flexion, abduction, and external rotation**

This maximizes capsular volume, thus lessening the pain



Proximal Interphalangeal Joint:
Swollen
Red
Circumferential

SEPTIC ARTHRITIS

Diagnosis:	-**Joint aspiration is definite test**. send for gram stain and culture. -Blood cultures must also be obtained -X-ray is not reliable in acute infection. -CT scan may help.

Infected synovial fluid is often turbid, has elevated PMNs, and has other characteristics that differentiate it from gout, pseudogout, and other inflammatory conditions

Aspirate is sent for gram stain and culture.

SEPTIC ARTHRITIS: SYNOVIAL FLUID ANALYSIS

	Normal	Abnormal
Diagnosis:	Highly viscous Clarity- transparent Essentially acellular Protein concentration about 1/3 of plasma Glucose concentration similar to blood Sterile	Less viscous Clarity- opaque WBC count usually >100,000cc (75% polys) May be lower with partially treated or low virulence organism Protein increased (not very specific) Glucose <25 mg/dl Culture often positive

SEPTIC ARTHRITIS

Treatments	
Orthosis	Splint during the acute phase, then introduce gentle PT to regain mobility -Affected joint is at risk for forming adhesions. Risk for adhesive caspulits.
Medications	IV antibiotics for 2-3 weeks: -Nafcillin or vancomycin for gram + cocci. -Third generation cephalosporin for gram – bacilli. -Ceftriaxone for suspected GC
Procedures	-Aspirate affected joint daily to remove necrotic material and follow serial WBC counts and cultures. -Arthroscopic and even open surgical debridement is often necessary, especially if there is resistance to IV antibiotic therapy

SEPTIC ARTHRITIS SUMMARY

-Usually via **Hematogenous Spread**

-In Children **, S. aureus** is the most common. H. influenzae used to be most common (consider if unvaccinated).

-In Adult: **S. aureus,** Strep , and gram-negative bacilli.

- Consider Gonococcal infection for both children and adults.

-Risk factors: Prosthetics, RA, DM, Infection, IV drugs, ETOH, >80 years old.

- Children and adult presents with constitutional symptoms.
- Infant: may limit movement of limbs usually hip will be in **Flexion, Abduction, and external rotation.**
- Older children and adults usually are effect in **the knee.**

-Diagnosis: **Joint Aspirations is definite** , may show elevated WBC, increased Protein, and low Glucose.

-Treatment: splint, gentle ROM, IV antibiotics (2-3 weeks): 3rd general cephalosporin (gram-) and vancomycin/nafcillin (gram+). Ceftriaxone for suspected GC. Daily aspiration and potentially surgery needed if resistant to medical management.

GONOCOCCAL ARTHRITIS

Epidemiology	-Mostly in adults but must also be considered in neonates and sexually active adolescents. -N. gonorrhoeae has accounted for as much as 70% of infectious arthritis in the <40 y.o. age group. -Women are 2-3x more likely than men to develop disseminated gonococcal infection (DGI) and arthritis. Higher risk during menses and pregnancy.
Causes Diagnosis	-Usual route is via asymptomatic mucosal colonization of the urethra, cervix, or pharynx. -Cultures from potentially infected mucosal sites.
Presentation	-Fever, chills, rash and articular symptoms. -Rash manifests as papules that progress to hemorrhagic pustules on the trunk and extensor surfaces. -Migratory arthritis of knees, hands, and feet is often prominent.
Treatment	ceftriaxone 1gm IV/IM daily , followed by an oral agent (i.e. doxycycline ~2 weeks).

Gonococcal septic arthritis is less common than DGI, but is always preceded by DGI
GC septic arthritis usually only involves a single joint, which will be full of pus
    Often does not reveal GC
    Obtain cultures from potentially infected mucosal sites

GAS GANGRENE

Clostridial Myonecrosis

Blisters
Erythema
“Crepitus” : bubble wrap.

GAS GANGRENE

Microorganism	**Clostridium perfingens:** anaerobic gram-positive spore-forming bacillus
Causes	-Typically begins after an open wound (intestinal surgery, burn injury, knife wound, prolonged rupture of membranes, intramuscular injection, black tar heroin injection (skin popping). -Incubation time usually 1-4 days.
Presentation	Earliest symptom is severe pain.* Rapidly progresses to localized tense edema, pallor, and tenderness. Often able to appreciate crepitus on palpation of the tissue. Skin typically becomes dark. Bullae develop. Thin, dirty brown, mousey-smelling discharge. Systemic toxicity: tachycardia, fever.

*Use osteopathic palpation

GAS GANGRENE

Diagnostics	X-ray shows gas in soft tissue CT with IV contrast will delineate Necrotizing Fasciitis. Gram stain may shows gram-positive or gram variable rods
Treatment	-Supportive therapy - Broad spectrum antibiotics. Penicillin G IV and clindamycin IV. -Hyperbaric oxygen - Immediate fasciotomy and debridement if compartment syndrome is present. -All cases ultimately will need surgery to resect the damaged tissue

*Use osteopathic palpation

NECROTIZING FASCIITIS

I. Severe, rare, rapidly progressive and potentially fatal.
Usually in extremities, perineum, abdominal wall.
Septic shock is common.

II.4 types:

- Type 1: Polymicrobial – 70-80% (bowel flora)
- Type 2: Monomicrobial – 20 – 30% (oral/skin flora)
- Type 3: Gram (-) – rare (marine flora)
- Type 4: Usually Trauma related in immunocompetent pts (fungal)

Up to 60% are group A Strep.

III. LRINEC Score for Necrotizing Soft Tissue Infection (https://www.mdcalc.com/calc/1734/lrinec-score-necrotizing-soft-tissue-infection) *
IV. Diagnostic: Labs (next slide).

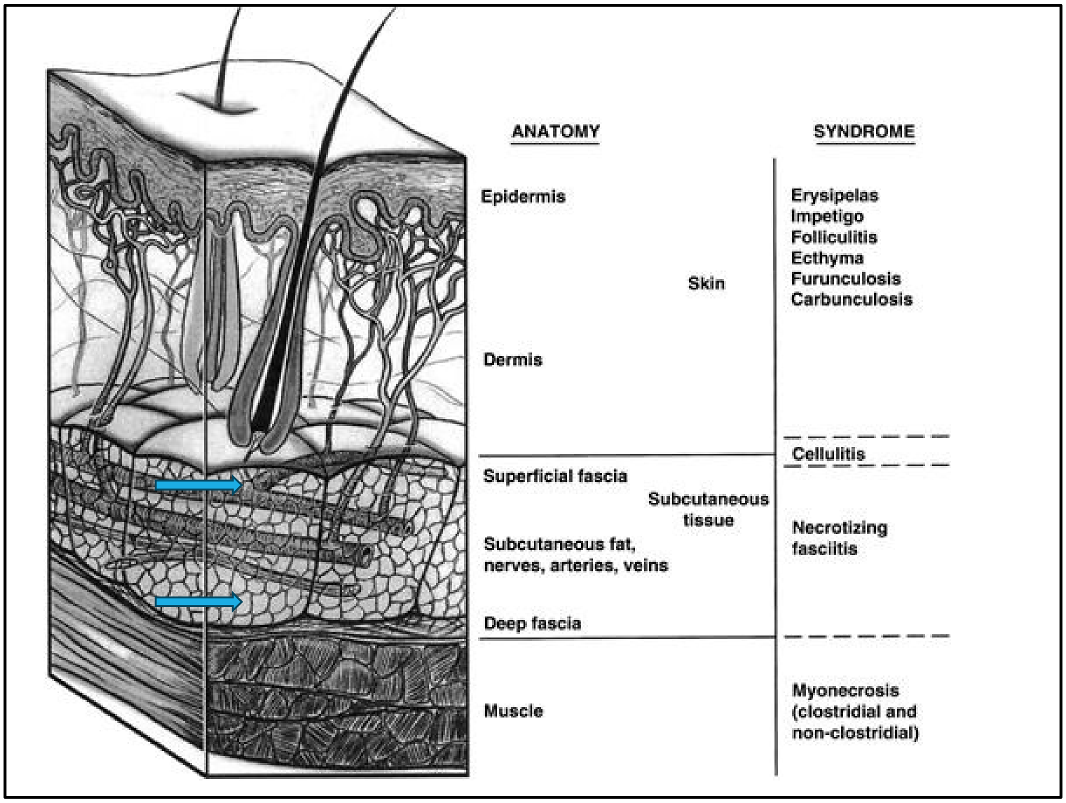
Imaging - CT scans demonstrate deep fascial thickening and enhancement, and the presence of fluid and gas within soft tissue planes in and around the superficial fascia. **

MRI is best BUT CT with IV contrast is faster, easier, and shows the above. ***

V. Treatment: Immediate Broad-Spectrum antibiotics, fluid resuscitation, to OR for debridement, Possible hemicorporectomy.

Other comorbidities worsen outcomes
*Laboratory Risk Indicator for Necrotizing Fasciitis score (LRINEC) is a simple tool used to support early diagnosis of Necrotizing Fasciitis (NF).
** if no superficial fascia involved and no enhancement: no NF.
*** should not send toxic patient to get one hour MRI .
****A hemicorporectomy is a complex surgical procedure that involves removing the lower half of the body, including the legs, genitalia, anus, rectum, pelvic bones, and urinary system

Laboratory Risk Indicator for Necrotizing Fasciitis		
CRP (mg/dL)	<15	0
	≥15	4
WBC (per mm³)	<15	0
	15-25	1
	>25	2
Hemoglobin (g/dL)	>13.5	0
	11-13.5	1
	<11	2
Sodium (mEq/L)	≥135	0
	<135	2
Creatinine (mg/dL)	≤1.6	0
	>1.6	2
Glucose (mg/dL)	≤180	0
	>180	1
Composite Score	**Score < 6**	**Low Risk**
	Score 6-7	**Intermediate**
	Score ≥ 8	**High Risk**



Infection in superficial fascia +enhancing in deep fascia+ Gas bubble in the skin: NF.

X-ray
Black spot: air from infection.
Feels like rice crispies.
Pain with palpation.

A. Air in SubC tissue going down to superficial fascial layer
B. Deep in muscle, migrating from hip up to pelvic and potentially to abdominal wall.

Dry Gangrene: Blacked area in extremities. Unsalvagable. OR for debridement and likely amputation.

Infections continues to move proximally.
Can lead to hip disaritulcatin.

GAS GANGRENE SUMMARY

- Gas Gangrene caused by **Clostridium perfingens**
- Usually after trauma.
-Presents with severe pain
- Physical exam shows dark skin and bullae.   Thin, dirty brown, mousey-smelling discharge. Crepitus on exam. ☆

- CT with IV contrast will delineate Necrotizing Fasciitis.
-Treatment: supportive therapy , Antibiotics (Penicillin G and clindamycin), Hyperbaric oxygen, and Immediate fasciotomy and debridement if compartment syndrome is present.

*Use osteopathic palpation

NECROTIZING FASCIITIS SUMMARY

-Severe, rare, rapidly progressive and potentially fatal.
- Usually in extremities, perineum, abdominal wall.

-**Up to 60% are group A Strep.**

-If suspicious for Necrotizing Fasciitis, can use **LRINEC (CRP, WBC, and CMP).**

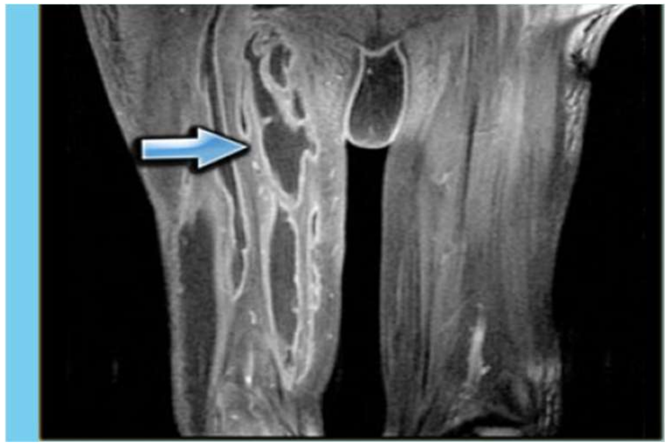
-X-ray can show gas and MRI is golden standard however CT with contrast is best in toxic patient .

-CT may show enhancement, and the presence of fluid and gas within soft tissue planes in and around the superficial fascia.

-Start on Immediate Broad-Spectrum antibiotics, fluid resuscitation, and consider OR for debridement.

-Poor prognosis.

*Use osteopathic palpation



PYOMYOSITIS: PURULENT INFECTION OF SKELETAL MUSCLE VIA HEMATOGENOUS SPREAD

Usually with abscess formation

PYOMYOSITIS

Epidemiology	Classic infection in the tropics but can occur in temperate regions Tropics: healthy individuals Temperate: Immunocompromised, trauma, IV drug use
Causes	**S. Aureus (MRSA)**
Presentation	Fever, pain and cramping localized to a single muscle group Often in lower extremities Stage 1: crampy local muscle pain, swelling, low-grade fever Stage 2: 10-21 days after onset. Fever, exquisite muscle tenderness, and edema. Frank abscess may be apparent Stage 3: systemic toxicity

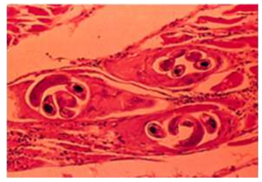
Circumferentially swollen
Tender and painful
Open wound on elbow.l

PYOMYOSITIS

Diagnostics	-Radiograph/CT with contrast or MRI if available -Cultures
Treatments	-Percutaneous drainage -Antibiotics usually Vancomycin +/- broad spectrum .

Time is Tissue
And Tissue is life.

TRICHINOSIS



Epidemiology	Common in Europe and US In US: 1/3 of cases attributed to consumption of **raw/undercooked** wild animal meat. ~1.5 million persons carry live Trichinella in musculature

Causes	• *Trichinella* Spiralis (Pork Worm)
	• Pathogenesis due to
	  ○ Host reaction to larval metabolites
	  ○ Lesions in striated muscle, heart and CNS
	  ○ Larvae invasion of muscle cells ~1 week after infection

-Found in many meat eating animals.
-Picture; *T. spiralis* encysted in muscle (looks like a bear paw).
-Symptoms depend on severity of infection (number of larvae).
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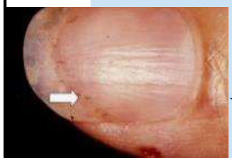
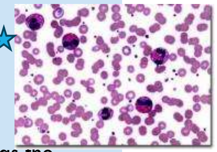
TRICHINOSIS





Presentation	Mild: or low # organism: subclinical. Heavy: large low # organism: ↓ immune response, rawness of meat Intestinal stage nausea, abdominal pain, diarrhea Muscle stage Muscle pain, tenderness, swelling, weakness Subungual splinter hemorrhages, eosinophilia
Diagnosis	ELISA is a primary screening test with Western blot used as the confirmatory assay from excretory/secretory antigens
Treatment	-Albendazole 400 mg orally twice a day for 8 to 14 days or mebendazole 200 to 400 mg orally 3 times a day for 3 days initially. Followed by 400 - 500 mg 3 times a day for 10 days can be used. -Analgesics may help relieve muscle pains. -For severe allergic manifestations or myocardial or central nervous system (CNS) involvement, prednisone 20 to 60 mg orally once a day is given for 3 or 4 days, then tapered over 10 to 14 days.

Efficacy uncertain.
Significant SE.

CYSTICERCOSIS

Epidemiology	Prevalent in areas with poor sanitation and pigs that have access to human
Causes	pork tapeworm Taenia solium
Presentation	Cysticerci in muscle: asymptomatic or nodules, inflammation Cysticerci in the eyes: loss of visual acuity Neurocysticercosis: seizures, and headaches
Diagnosis (Difficult)	-CT head (IV contrast best). -Intramuscular cyst can't be seen under they undergo calcification on imaging. -Antibody detection via blood testing.
Treatment	**Albendazole or praziquantel** Steroids to decrease inflammation Surgical removal of cysts

VIRAL INFECTIONS OF THE MUSCULOSKELETAL SYSTEM:

Common pediatric rash causing pathogens	
Rubella virus: 	RNA virus; (German measles). Also a teratogen for fetuses, this can cause arthritis in up to 30% of adult female patients.
Parvovirus B19: 	DNA virus ("slapped cheek syndrome"); causes arthritis in 60% of infected adults, esp. females.

Rubella: consider booster.

VIRAL INFECTIONS OF THE MUSCULOSKELETAL SYSTEM:

RNA arboviruses (mosquito-borne)	Now endemic in the western hemisphere
Chikungunya virus	Sudden, acute illness with high fever (>102 F), headache, intense fatigue, back pain, maculopapular rash, and incapacitating polyarthralgias or polyarthritis (with joint swelling/tenderness).
Dengue virus	Similar to Chikungunya with complaints of rapid onset of high fever, headache, diffuse and severe body pain (both muscle and bone), retro-orbital pain, prostration, maculopapular rash, capillary friability, nausea, and vomiting.
Zika virus:	Similar to Chikungunya, but clinically, only 20% of those infected become symptomatic, usually with a low-grade febrile illness with rash, rigors, polyarthralgia, and/or conjunctivitis and eye pain. Headache and myalgia are also common.

TETANUS

Epidemiology	-Rare in the US. -Most cases occur among older adults who are unvaccinated or **inadequately vaccinated.**
Causes Transmission	Caused by **Clostridium tetani** Spores (viable for many years) Ubiquitous: soil, dust, animal feces Puncture wound/trauma Requires low tissue oxygenation Obligate anaerobe

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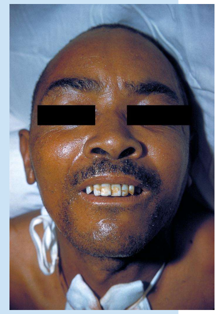
Foot/hands. Worsened by PVD.

TETANUS

Presentation	Spastic paralysis = strong muscle spasms <u>Early symptoms</u> Lockjaw (trismus) = spasms of masseter muscles Risus sardonicus = grimace Difficulty swallowing = pharyngeal muscle spasms <u>Late findings</u> Muscle rigidity Opisthotonus = arching of back (not subtle!).



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No or minimal inflation at the site of infection .

CLINICAL PRESENTATION

Generalized Tetanus

Exaggerated reflexes, muscle rigidity, uncontrolled muscles spasms

Generalized rigidity

As the disease progress the spasms become more intense and frequent ➡ apnea, fractures, dislocations

Death from respiratory failure

Neonatal Tetanus

Inability to suck 3-10 days after birth

Irritability, excessive crying, grimaces, intense rigidity

This baby cannot breast feed or open his mouth because the muscles in his face have become too tight

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Generalized tetanus is the most common form of tetanus, occurring in approximately 80% of cases. Patients present with a descending pattern of muscle spasms, first presenting with lockjaw, and risus sardonicus (rigid smile because of sustained contraction of facial musculature). This can progress to a stiff neck, difficulty swallowing, and rigid pectoral and calf muscles. These spasms can occur for up to 4 weeks, with full recovery taking months. Autonomic instability can also occur in these patients with fever, dysrhythmia, labile blood pressure and heart rate, respiratory difficulties, catecholamine excretion, and even early death.

Neonatal tetanus is a generalized form of tetanus that occurs in newborns of unimmunized mothers or from infection through a contaminated instrument when cutting the umbilical cord. Infants of immunized mothers generally do not get tetanus due to passive immunity from the mother. Those who are infected, exhibit irritability, poor feeding, facial grimacing, rigidity, and severe spastic contractions triggered by touch. There have been case reports of long-term consequences in survivors of neurodevelopmental impairments, behavioral problems, and deficits in gross motor, speech, and language development.

Opposite is neonatal botulism.

TETANUS

Diagnostics	Clinically based Gram stain and culture often negative
Treatment	-Early wound treatment: Debridement of wounds to remove organisms and create an aerobic environment -Antibiotic Therapy: Metronidazole or penicillin To eliminate bacteria -> reduce toxin production -Neutralization of unbound toxins – IM or IV human tetanus immune globulin. -Control of muscle spasms Spasmolytic drugs, sedation, muscle relaxants, respiratory support, etc. -Vaccination (toxoid: Td, DTaP, DT) Infection does not confer immunity, administer immediately after diagnosis

Picture: Terminal spore in GPR
Drumstick appearance
(tennis racket)

- Vaccination will not give immunity for 2 weeks: Have to give both IM and vaccine.

SUMMARY (SLIDE 52-65)

-Pyomyositis: infection via hematogenous spared in skeletal muscle (usually thigh). Usually in Tropics caused by S. Auerus (MRSA). Antibiotics + percutaneous drainage.
-Trichinosis: uncooked meat. *Trichinella Spiralis* cyst in (bear paw). Subungual splinter hemorrhages and eosinophilia. Tx w/ Albendazole or mebendazole.
-**Cysticercosis:** *Taenia solium.* Neurocysticercosis can present with seizures and headaches. Dx with CTH. Tx w/ Albendazole or praziquantel.
Rubella: RNA virus: arthritis in 30% of female adult. May need booster vaccine.
Parvovirus B19: DNA virus: slapped check in 60% of adults.
Chikungunya virus: high fever, maculopapular rash, polyarthralgia
Dengue virus: myalgia, retro-orbital pain, capillary friability
Zika virus: only 20% symptomatic. Low fever, rash, polyarthralgia's, and conjunctivitis and eye pain.
Tetanus: Generalized tetanus most common (80%). Presents with spastic paralysis, trismus, risus sardonicus, and opisthotonus. Tx w/ wound debridement, antibiotics (Metronidazole or penicillin), symptoms control, **human tetanus immune globulin AND vaccination.**

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capillary friability: bruises, bleeding.

FINAL PEARLS

-Treat with broad spectrum even if initial diagnostics unremarkable.

-S. Auerus usually the most common, but consider uniqueness of special population (immunocompromised, IV drug uses, vascular insufficient) .

-CT maybe more practical than MRI : **Time is Tissue and Tissue is life.**

-Children usually have more constitutional symptoms and treatment is more curative.

- Inspection and palpate without clothes/shoe/socks.

QUESTIONS?

Thank you!

THANKS/REFERENCES:

Necrotizing fasciitis
Pejman Davoudian, FRCA EDIC FFICM, Neil J Flint, FRCA EDIC
Continuing Education in Anaesthesia Critical Care & Pain, Volume 12, Issue 5, October 2012, Pages 245–250, https://doi.org/10.1093/bjaceaccp/mks033
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